

LINCOLN 11 SW, NE, USA

WMO: 744410

Lat: 40.6954N

Lon: 96.8541W

Elev: 418

StdP: 96.40

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 94996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-16.1	-23.5	0.5	-17.8	-21.0	0.6	-15.7	8.4	-0.3	7.8	-0.9	2.3	N/A	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	34.8	23.3	33.1	23.2	31.5	22.9	25.6	31.5	24.7	30.5	23.8	29.4	3.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.8	29.6	22.9	18.6	28.4	22.0	17.5	27.5	81.3	31.5	77.1	30.4	73.4	29.4	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.7	6.0	-23.1	37.4	1.6	1.6	-24.2	38.6	-25.2	39.5	-26.1	40.5	-27.3	41.7		
(5)			-23.3	26.9	1.9	0.9	-24.7	27.5	-25.8	28.0	-26.9	28.5	-28.2	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-3.6	-2.1	5.5	11.1	16.7	22.4	24.7	23.3	19.5	12.0	4.9	-1.8	(6)
(7)		DBStd	11.33	6.88	6.89	6.75	5.65	4.83	3.62	3.40	3.26	4.60	5.46	5.88	6.10	(7)
(8)		HDD10.0	1575	423	341	172	54	6	0	0	0	1	42	172	365	(8)
(9)		HDD18.3	3255	681	572	399	224	90	7	2	3	41	208	404	623	(9)
(10)		CDD10.0	1978	0	2	33	87	213	371	454	413	284	102	18	1	(10)
(11)		CDD18.3	616	0	0	1	7	39	128	198	158	74	11	0	0	(11)
(12)		CDH23.3	6181	0	0	23	96	401	1268	2057	1497	723	111	5	0	(12)
(13)	CDH26.7	2424	0	0	4	23	119	474	905	607	263	28	1	0	(13)	
(14)	Wind	WSAvg	2.8	3.0	3.1	3.4	3.5	2.9	2.5	2.2	2.1	2.5	2.7	2.9	2.8	(14)
(15)	Precipitation	PrecAvg	730	17	21	42	65	119	114	107	80	71	53	37	20	(15)
(16)		PrecMax	1074	33	61	93	112	269	219	307	171	125	132	80	66	(16)
(17)		PrecMin	538	1	2	0	16	30	5	25	26	11	3	3	1	(17)
(18)		PrecStd	139	9	15	26	22	69	47	65	39	34	35	23	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	20.0	26.0	29.7	32.7	35.1	37.8	36.6	34.4	30.2	23.4	15.4	(19)
MCWB			7.5	9.5	15.0	18.4	19.9	23.7	22.5	22.5	21.8	18.2	13.4	8.8	(20)	
2%		DB	11.3	14.5	22.7	26.1	29.6	33.0	35.2	34.1	31.9	26.1	19.6	12.1	(21)	
		MCWB	5.3	8.7	12.7	16.0	19.6	23.4	23.9	22.8	21.9	16.2	12.1	6.6	(22)	
5%		DB	8.2	10.7	19.5	23.3	27.7	31.2	33.4	31.9	30.0	23.8	16.6	9.5	(23)	
		MCWB	3.8	5.6	11.7	14.2	18.8	22.9	24.4	23.1	21.2	15.0	10.2	5.3	(24)	
10%		DB	5.5	7.5	16.2	20.7	25.5	29.5	31.4	30.1	27.5	21.1	13.8	6.7	(25)	
		MCWB	2.4	4.0	10.2	12.9	17.7	22.0	24.1	22.4	19.9	13.7	8.6	3.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	12.5	17.5	18.4	22.7	26.0	27.1	26.4	24.0	20.2	15.4	11.8	(27)
MCD B			13.7	16.5	24.8	25.7	28.7	32.1	32.6	32.3	31.4	25.2	19.4	14.1	(28)	
2%		WB	5.8	9.0	15.2	16.8	21.1	24.8	25.8	25.1	23.0	17.9	12.9	7.8	(29)	
		MCD B	10.2	15.4	20.2	23.3	27.0	30.8	31.7	30.9	29.7	23.3	17.9	11.4	(30)	
5%		WB	4.1	5.9	12.7	15.4	20.2	23.9	25.1	24.0	22.2	16.3	11.0	5.5	(31)	
		MCD B	7.6	10.5	18.8	20.8	26.6	29.6	30.9	29.5	28.2	21.6	15.6	9.5	(32)	
10%		WB	2.4	3.9	10.6	13.8	19.0	22.9	24.1	23.0	21.3	14.7	9.0	3.6	(33)	
		MCD B	5.2	7.3	16.5	19.0	24.7	28.4	29.5	28.1	26.3	19.4	13.6	6.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.1	10.6	12.2	12.3	11.6	11.1	11.2	11.7	12.5	12.3	11.5	9.8	(35)	
5% DB		MCD BR	14.3	15.6	16.7	16.2	14.3	12.4	13.0	13.6	13.8	15.3	15.2	13.9	(36)	
		MCWBR	9.4	9.9	9.4	8.5	6.2	6.0	5.3	5.6	5.9	8.0	8.6	9.3	(37)	
		MCD BR	13.2	14.9	15.1	13.4	12.3	11.5	10.8	11.5	11.6	13.0	13.7	13.1	(38)	
5% WB		MCWBR	9.0	9.8	8.6	7.9	6.1	6.4	5.6	5.7	5.7	7.7	8.5	9.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.281	0.298	0.322	0.374	0.402	0.412	0.419	0.418	0.379	0.331	0.293	0.278	(40)
taud		2.475	2.445	2.417	2.294	2.286	2.314	2.293	2.300	2.359	2.481	2.513	2.486	(41)		
Ebn at Noon		890	922	932	898	877	865	854	845	858	870	868	863	(42)		
Edh at Noon		76	90	104	127	131	128	129	125	110	87	72	69	(43)		
(44)	All-Sky Solar Radiation	RadAvg		2.14	3.02	3.98	5.03	5.90	6.63	6.59	5.75	4.71	3.37	2.38	1.81	(44)
(45)		RadStd		0.18	0.19	0.33	0.33	0.43	0.44	0.40	0.30	0.31	0.36	0.17	0.17	(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page